

A system has RAM which can store four pages simultaneously to satisfy page requests. Write a program to calculate percentage page hit and page fault if the a system uses Optimal page replacement technique for the page references entered by the user i.e. 1, 2,3,4, 1,2,5, 1,2,3,4, 5 (No of Frames=3)'

Code :

```
#include <stdio.h>

int main()
{
    int pages[50], frames[10];
    int n, f, i, j, k;
    int faults = 0, hits = 0;

    printf("Enter number of pages: ");
    scanf("%d",&n);

    printf("Enter page reference string:\n");
    for(i=0;i<n;i++)
        scanf("%d",&pages[i]);

    printf("Enter number of frames: ");
    scanf("%d",&f);

    for(i=0;i<f;i++)
        frames[i] = -1;

    for(i=0;i<n;i++)
    {
        int flag = 0;

        for(j=0;j<f;j++)
        {
            if(frames[j] == pages[i])
            {
                hits++;
                flag = 1;
                break;
            }
        }
    }
```

```
if(flag == 0)
{
    int farthest = -1, pos = -1;

    for(j=0;j<f;j++)
    {
        int found = 0;

        for(k=i+1;k<n;k++)
        {
            if(frames[j] == pages[k])
            {
                if(k > farthest)
                {
                    farthest = k;
                    pos = j;
                }
                found = 1;
                break;
            }
        }

        if(found == 0)
        {
            pos = j;
            break;
        }
    }

    frames[pos] = pages[i];
    faults++;
}

printf("\nPage Faults = %d", faults);
printf("\nPage Hits = %d", hits);

printf("\nPage Fault Percentage = %.2f%%", (float)faults/n*100);
printf("\nPage Hit Percentage = %.2f%%", (float)hits/n*100);

return 0;
}
```

Output:

```
Enter page reference string:
```

```
1 2 3 4 1 2 5 1 2 3 4 5
```

```
Enter number of frames: 3
```

```
Page Faults = 7
```

```
Page Hits = 5
```

```
Page Fault Percentage = 58.33%
```

```
Page Hit Percentage = 41.67%
```